



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)

Re-Accredited by NAAC with 'A++' Grade

ANISH NAIR

BATCH: A1

PRN:25070122037

Subject: Programming with Java

ASSIGNMENT No: 4

TITLE: Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

Problem Statement

Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

Learning Objectives

To perform conversion between primitive data types and wrapper objects using Java wrapper classes.

Theory

Wrapper classes are predefined classes that encapsulate primitive data types into objects. Java provides wrapper classes such as Integer, Double, Character, and Boolean for every primitive type. They support autoboxing, unboxing, parsing, and conversion methods. Wrapper classes are widely used in Java Collections Framework and when object representation of primitive values is required.

Outcome

```

1  import java.util.Scanner;
2
3  public class StudentMarksDemo {
4      public static void main(String[] args) {
5          Scanner sc = new Scanner(System.in);
6
7          String[] marksStr = new String[3];
8          System.out.println("Enter marks for 3 subjects (as String input):");
9          for (int i = 0; i < 3; i++) {
10             System.out.print("Subject " + (i + 1) + ": ");
11             marksStr[i] = sc.nextLine();
12         }
13
14         Integer total = 0;
15         for (String mark : marksStr) {
16             Integer marksInt = Integer.parseInt(mark);
17             total += marksInt;
18         }
19
20         System.out.println("Total Marks: " + total);
21         System.out.println("Average Marks: " + (total / 3.0));
22     }
23 }

```

```

Enter marks for 3 subjects (as String input):
Subject 1: 85
Subject 2: 90
Subject 3: 78
Total Marks: 253
Average Marks: 84.33333333333333

```

```

1  import java.util.Scanner;
2
3  public class EmployeePayrollDemo {
4      public static void main(String[] args) {
5          Scanner sc = new Scanner(System.in);
6
7          System.out.print("Enter Employee ID: ");
8          Integer empId = Integer.parseInt(sc.nextLine());
9
10         System.out.print("Enter Basic Salary: ");
11         Double basicSalary = Double.parseDouble(sc.nextLine());
12
13         System.out.print("Enter Bonus Amount: ");
14         Double bonus = Double.parseDouble(sc.nextLine());
15
16         Boolean isValidSalary = basicSalary > 0;
17
18         if (isValidSalary) {
19             Double netSalary = basicSalary + bonus;
20             System.out.println("\nEmployee ID: " + empId);
21             System.out.println("Basic Salary: Rs. " + basicSalary);
22             System.out.println("Bonus: Rs. " + bonus);
23             System.out.println("Net Salary: Rs. " + netSalary);
24         } else {
25             System.out.println("Invalid Salary Entered. Cannot Process Payroll.");
26         }
27     }
28 }

```

```

Enter Employee ID: 101
Enter Basic Salary: 35000
Enter Bonus Amount: 5000

```

```

Employee ID: 101
Basic Salary: Rs. 35000.0
Bonus: Rs. 5000.0
Net Salary: Rs. 40000.0

```

Conclusion: Wrapper classes let primitive values (parsed from String input) be treated as objects, enabling parsing, validation, and arithmetic operations needed for tasks like marks totalling and payroll calculation.

Exercise

1. Create a program to convert student marks from String format to Integer and calculate total marks.
2. Develop an **Employee Payroll System** that accepts employee IDs, basic salary, and bonus amounts from the user. Convert the entered values into wrapper objects and perform validation operations to ensure valid salary values before calculating the net salary.